



Early-life menstrual characteristics and gestational diabetes in a large US cohort

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Abstract

Background: Associations between early-life menstrual cycle characteristics (MCC) and gestational diabetes (GDM) remain unclear.

Objectives: To evaluate associations between early-life MCCs and GDM in first pregnancy, across pregnancies and its recurrence.

Methods: This analysis included participants from a US-based digital cohort enrolled between 11/2019 and 9/2023 who provided consent, completed relevant surveys, were without diabetes and aged ≥ 18 at first pregnancy ($n = 30,473$). Age at menarche [< 11 (early), 11–15 (referent), ≥ 16 (late) years] and time from menarche to cycle regularity [< 1 (referent), 1–2, 3–4, ≥ 5 years, not yet regular, regular after hormones] were self-recalled at enrolment. Additionally, the last three categories were considered prolonged time-to-regularity (PTTR). GDM history was recalled at enrolment for each pregnancy. We restricted to pregnancies of ≥ 24 weeks with a live birth. We evaluated associations of early-life MCCs with GDM at first pregnancy using modified Poisson regression, across pregnancies using cluster-weighted Poisson generalised estimating equation and GDM recurrence using multinomial logistic regression, adjusted for sociodemographic, early-life factors and age at pregnancy. Missing variables were imputed with multiple imputation by chained equations.

Results: Among 30,473 participants, 20,591 had eligible first pregnancies, of which 5.9% reported GDM. In 17,512 participants with ≥ 2 pregnancies, 8.3% had GDM once and 3.7% had recurrent GDM. Early menarche (< 11 years, vs. 11–15 years) was associated with GDM in first pregnancy (RR 1.34, 95% CI 1.15, 1.57), across pregnancies (RR 1.24, 95% CI 1.10, 1.39) and recurrence (OR 1.51, 95% CI 1.21, 1.89). PTTR was associated with GDM in the first pregnancy (RR 1.22, 95% CI 1.08, 1.38), across pregnancies (RR 1.16, 95% CI 1.05, 1.27) and recurrence (OR 1.19, 95% CI 0.99, 1.43).

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Conclusions: Earlier menarche and prolonged time-to-regularity are associated with higher risk of GDM and recurrence, suggesting menstrual characteristics during childhood/adolescence as potential early-life markers for GDM.

KEYWORDS

digital cohort study, gestational diabetes, irregular cycles, menarche, menstrual cycle, recurrent gestational diabetes

1 | BACKGROUND

Gestational diabetes mellitus (GDM) complicates 2–10% of pregnancies in the United States¹ Prevalence is increasing,² raising concerns regarding pregnancy-related and chronic health risk.³ Established risk factors like advanced age, obesity, family history of diabetes and lifestyle factors⁴ during the perinatal window have been predominantly studied. Investigating markers in early-life, specifically during puberty, a physiological transition of considerable metabolic and hormonal change including increased insulin resistance, altered body composition and lipid profiles,⁵ may offer insights for early screening. Previous GDM is a strong risk factor for subsequent GDM.⁶ Identifying additional markers for recurrent GDM is also clinically important, but evidence remains limited.⁷ Examination of early-life markers, such as menstrual characteristics, may provide further understanding of GDM and its recurrence.

Menarche and time to establishing menstrual regularity are two hallmarks of early-life menstrual health.⁸ Increased adiposity has been associated with early menarche and insulin resistance.⁹ Previous studies with distinct demographic composition (e.g. an occupation-derived cohort in the United States, and other pregnancy cohorts in Australia, United Kingdom or China) either suggested a positive association between earlier menarche and GDM^{10–14} or found no association.¹⁵ Most studies were moderate in size or from single-pregnancy cohorts, limiting evidence on GDM risk across multiple pregnancies. The menarche-to-regularity interval signifies reproductive axis maturation.¹⁶ While some studies found associations between long or irregular cycles in adulthood and higher GDM risk,^{15,17} none have examined the impact on GDM by time-to-regularity—an important yet understudied marker of early-life menstrual health.

In this study, we examined the association of age at menarche and time-to-regularity with GDM in first pregnancy, across pregnancies and recurrence in a large US digital cohort (conceptual model in Figure S1).

2 | METHODS

2.1 | Study design and cohort selection

The Apple Women's Health Study is a prospective digital cohort study in the United States, designed to gain better understanding of the relationship among menstrual cycles, health and

Synopsis

Study question

Could early-life menstrual characteristics, including age at menarche and time from menarche to establishing regular cycles (time-to-regularity), be markers for gestational diabetes (GDM) and its recurrence?

What is already known

Scant studies, mostly moderate in size or from single-pregnancy cohorts, suggest an association between earlier menarche and GDM. There was no existing study reported on time-to-regularity and GDM or its recurrence. Large cohorts with heterogeneous demographics and information of multiple pregnancies are needed to evaluate these associations.

What this study adds

We found that earlier menarche and prolonged time from menarche to cycle regularity are associated with higher risk of GDM and its recurrence, suggesting early-life windows (such as childhood and adolescence) may provide insight in early markers for GDM.

behaviour.¹⁸ Eligible participants were users of the Apple Research app on their iPhone (sole usage, including their iCloud account) who have ever menstruated, live in the United States, were at least 18 (19 in Alabama and Nebraska, 21 in Puerto Rico) years old and able to communicate in English. Enrolment began November 2019 and is ongoing. Written informed consent of participation was provided at enrolment. AWHS study design, eligibility and data were described previously.¹⁸ For this analysis, we included participants who enrolled between 11/2019 and 9/2023 and responded to the Reproductive History Survey at enrolment. We further restricted to participants who indicated ever being pregnant, agreed to provide pregnancy history information and responded to questions of demographics, medical history and pregnancy complications. We excluded those with self-reported type 1 or 2 diabetes or did not provide age at menarche information. To limit potential

bias from pregnancies occurring soon after menarche (that accurate time-to-pregnancy information may be unavailable),¹⁹ we further restricted analyses to participants who reported ≥ 18 years old at first pregnancy. The final study population included 30,473 participants; specific data analysis was limited to eligible subsets (flowchart in Figure S2).

2.2 | Exposures

We identified early-life menstrual characteristics from survey questions. First, we asked the participants 'At what age did you have your first menstrual period? It's okay to estimate', with the options: '7 years old or younger'; integers from 8 to 15; '16 years old or older'; 'I don't know'; or 'I prefer not to answer'. We derived: (1) age at menarche (years): of note, we assigned the value of 7 to '7 years old or younger' (0.3%) and the value of 16 to '16 years old or older' (3.4%); (2) age at menarche category (with commonly used cutoffs): early (<11), late (≥ 16) or 11–15 years (referent group).²⁰ Then, participants were asked: 'After your first menstrual cycle, how long did it take for your cycle to become regular?' with the following options: 'Less than 1 year', '1–2 years', '3–4 years', 'More than 5 years', 'After using hormones (e.g. birth control pills)', 'They're not yet regular', 'I don't know' or 'I prefer not to answer'. We derived: (1) prolonged time-to-regularity (yes/no), where 'yes' referred to those who took >5 years or used hormones to establish regularity, or never established regularity; (2) time-to-regularity (original categories, <1 year as referent group). For all questions, don't know/prefer not to answer/no-response were considered missing.

2.3 | Outcomes

For each reported pregnancy, participants were asked the question 'Did you have any complications related to this pregnancy? (select all that apply)' with 'gestational diabetes (diabetes only during pregnancy)' as the first option among the list of options (Table S1). Those who selected GDM were counted as having GDM, and those who selected other complication(s) but not GDM, or selected 'none of the above' were considered not having GDM. As screening for GDM generally occurs during 24–28 weeks of gestation,²¹ and potential bias may arise from pregnancy loss or stillbirth,²² our primary analyses were restricted to pregnancies that lasted ≥ 24 weeks (from self-reported gestational length) and ended with a live birth (from self-reported pregnancy outcome, Table S1). We refer to these pregnancies as 'eligible' pregnancies. We derived: (1) GDM (yes/no) in first pregnancy (restricted to eligible gravida 1, $n=20,591$); (2) GDM (yes/no) in each of the reported pregnancies, among 26,091 participants with ≥ 1 eligible pregnancies; (3) GDM occurrence categories, among 17,512 participants with ≥ 2 eligible pregnancies: no GDM (did not ever report GDM), GDM single

occurrence (reported GDM only in one of the pregnancies) or GDM recurrence (reported GDM in ≥ 2 pregnancies).

2.4 | Covariates

We considered the following self-reported/self-identified variables as potential confounders. (1) Race and ethnicity: Asian, Hispanic, non-Hispanic Black, non-Hispanic White, multiple races (self-identified >1 option) and we combined into 'other races' several infrequently chosen groups (American Indian or Alaska Native/Middle Eastern or North African/Native Hawaiian or Pacific Islander/ 'None of these fully describe me'). Race and ethnicity was included because of its known associations with both menarche and GDM.^{23,24} (2) Socioeconomic status (SES): self-perceived score using the MacArthur Scale of Subjective Social Status²⁵ (categorised as 0–3 [low]/4–5 [medium]/6–9 [high]) reported at enrolment, used as a surrogate for early-life SES. (3) Information on participant's own preterm birth status and birthweight (known associations with age at menarche)²⁶: classified as born preterm, term with low birthweight (<2500 g), term with normal birthweight (2500–4000 g) or term with macrosomia (>4000 g).²⁷ (4) Family history of diabetes or other metabolic conditions (obesity/high cholesterol/hypertension).

We considered self-reported age at pregnancy as a strong risk factor for GDM, and further adjusted for this variable in a separate model. Previous studies suggested that earlier menarche was associated with younger age at first pregnancy mediated by earlier sexual intercourse,²⁸ while younger age is associated with lower GDM risk.²⁹ The age-adjusted results may further elucidate the exposure-outcome associations, isolating them from the influence of behavioural, psychosocial or biological factors related to age at pregnancy.

Among 7614 participants who responded to an additional, one-time survey (released 8/2023 to all existing participants) on self-recalled weight and height at menarche, we derived the BMI-for-age z-score at menarche following the Centers for Disease Control and Prevention (CDC) Growth Chart.³⁰ The scoring is a valid method to construct BMI for those aged 2–20 years.³¹ We categorised BMI at menarche (underweight/healthy/overweight/obesity, using z-score cutoffs from CDC) as an additional confounder.^{32,33}

We did not adjust for variables such as pre-pregnancy BMI, dietary factors, polycystic ovary syndrome (PCOS) or other metabolic disorders developed in adulthood, because they may be on the exposure-outcome pathway as potential mediators with inherent biological mechanisms.^{34–37} Adjusting for these factors may partially block the associations from which menstrual cycle characteristics in childhood/adolescence could serve as an early-life marker of GDM. Understanding the complexity of potential mediating factors is a next step and is beyond the scope of this analysis. Furthermore, if unmeasured confounders exist, adjusting for these potential

mediators may further introduce collider stratification bias, as illustrated in [Figure S1](#).

2.5 | Statistical analysis

We performed the following statistical analyses. First, we summarised the distributions of participant characteristics, overall and stratified by age at menarche category. Then, we used different statistical models to evaluate the associations between each menstrual characteristic and the outcomes. (1) To estimate the relative risk (RR) of GDM in first pregnancy, we used modified Poisson regression models.³⁸ We additionally used Poisson generalised additive models (GAMs) with cubic regression splines to allow for potential non-linear associations. (2) To estimate the relative risk of GDM across all eligible pregnancies, we used cluster-weighted Poisson generalised estimating equations (CW-GEE) with robust standard error estimation, a method that accounts for within-individual correlations of the GDM outcome (previous GDM is likely correlated with future GDM within an individual with two or more pregnancies) and applied a weighted approach to account for informative cluster size.^{39,40} Grand multiparity (e.g. parity >5) is indicative of higher risk of pregnancy complications and adverse outcomes.⁴¹ In this analysis, the weights in CW-GEE were calculated as the inverse of number of pregnancies for each individual to address the potential bias from informative cluster size. (3) For GDM occurrence category among those with ≥ 2 eligible pregnancies (GDM single occurrence or recurrence, vs. no GDM), we used multinomial logistic regression models. We evaluated the associations in unadjusted models, models adjusted for confounders and models further adjusted for age at pregnancy [for (1) and (3), we adjusted for age at first pregnancy; for (2), we adjusted for age at each pregnancy].

Among the 7614 participants with BMI at menarche, we further adjusted for BMI at menarche categories for the associations between the menstrual characteristics and GDM in the first pregnancy. We also used multiple imputation (described below) to impute BMI at menarche among all participants. Additionally, to account for possible selection bias arising from non-response to ad hoc survey (considered as lost-to-follow-up), we modelled the probability of lost-to-follow-up to derive inverse probability weights (IPW).⁴² We used the above-mentioned covariates (sociodemographic and health characteristics) to model the probability of loss-to-follow-up and generated weights to be applied to the regression models; these weighted estimates correspond to the associations that would have been observed had everyone provided BMI at menarche information. We compared the BMI-adjusted estimates across these three approaches.

2.6 | Missing data

The missingness for variables ranged between 0% and 1.4% for age at pregnancy or GDM, 0.2–3% for race and ethnicity, SES or

family history information, 16–18% for participant's recalled own preterm birth status and birthweight, and around 14% for time to regularity (details in [Figure S3](#)). For the main models (modified Poisson regression, CW-GEE and multinomial logistic regression), we imputed these missing variables using multiple imputation chained equations (MICE) with 50 imputed datasets, based on the exposures, outcomes and covariates values. RR estimates from the 50 imputed datasets were combined and averaged using Rubin's rules.⁴³

2.7 | Sensitivity analysis

We performed sensitivity analyses to evaluate the robustness of our main findings. Given the temporal changes in GDM screening and diagnostic criteria in the US⁴⁴ and in early-life menstrual characteristics,²⁰ we further adjusted for calendar year of pregnancy. We also expanded to pregnancies regardless of gestational length or pregnancy outcome.^{45,46} Third, we restricted our analyses to singleton pregnancies. Fourth, we applied mixed effects logistic regression (MELR) models as an alternative approach to account for within-individual correlations for GDM across all eligible pregnancies. We also limited information to up to the 4th pregnancy per individual as an alternative way to account for potential bias by grand multiparity. We tested for effect modification (EM) by self-reported polycystic ovary syndrome (PCOS) given its known association with early or late menarche, irregular cycles and GDM risk.⁴⁷ EM was evaluated on the additive scale using relative excess risk due to interaction (RERI).⁴⁸ We also evaluated GDM in subsequent pregnancies among those ($n=2112$) who ever had GDM.

Analyses were conducted in Python (version 3.6) and R (version 4.1.2).

2.8 | Ethics approval

This study was approved by the Institutional Review Board (IRB) at Advarra (CIRB #PRO00037562).

3 | RESULTS

Among the 30,473 participants, 20,591 had a first pregnancy resulted in live birth at ≥ 24 weeks. Of those eligible first pregnancies, 5.9% reported GDM. There were 26,091 participants with ≥ 1 eligible pregnancy (median = 2; range: 1–10). Among 17,512 participants who had ≥ 2 eligible pregnancies, 8.3% had GDM single occurrence and 3.7% had recurrent GDM overall (53.6% among those with GDM in first pregnancy). Individuals with early menarche were more likely to be younger at enrolment, self-identify as non-White, of low SES, born preterm and younger at first pregnancy ([Table 1](#)). More than half of the participants reported time-to-regularity ≤ 2 years ([Table 1](#)).

TABLE 1 Characteristics of the 30,473 AWHs participants, overall and by age at menarche category.

Baseline characteristics ^a	Overall	By age at menarche, years old			Those whose 1st pregnancy was ≥ 24 weeks with a live birth ^b
		<11	11–15	≥ 16	
N	30,473	3584	25,695	1194	20,591
Age at enrolment					
Mean $\pm$ SD	39.8 $\pm$ 10.4	37.2 $\pm$ 10.0	40.1 $\pm$ 10.3	42.5 $\pm$ 11.0	40.4 $\pm$ 10.1
Median (IQR)	39 (33–46)	36 (30–43)	39 (33–46)	41 (34–49)	39 (33–46)
Race and ethnicity, n (%)					
Asian	618 (2.0)	78 (2.2)	526 (2.0)	14 (1.2)	383 (1.9)
Hispanic	1917 (6.3)	346 (9.7)	1521 (5.9)	50 (4.2)	1264 (6.1)
Multiple races	2756 (9.0)	404 (11.3)	2267 (8.8)	85 (7.1)	1754 (8.5)
Non-Hispanic Black	1494 (4.9)	299 (8.3)	1134 (4.4)	61 (5.1)	1011 (4.9)
Non-Hispanic White	23,007 (75.5)	2357 (65.8)	19,702 (76.7)	948 (79.4)	15,735 (76.4)
Other races ^c	596 (2.0)	84 (2.3)	483 (1.9)	29 (2.4)	382 (1.9)
SES at enrolment, n (%)					
Low (0–3)	7671 (25.2)	1196 (33.4)	6215 (24.2)	260 (21.8)	5132 (24.9)
Medium (4–5)	12,434 (40.8)	1501 (41.9)	10,449 (40.7)	484 (40.5)	8394 (40.8)
High (6–9)	10,315 (33.8)	876 (24.4)	8993 (35.0)	446 (37.4)	7022 (34.1)
Participant's birthweight and preterm status, n (%)					
Born preterm	2609 (8.6)	377 (10.5)	2120 (8.3)	112 (9.4)	1663 (8.1)
Born term, with low birthweight (<2500 g)	995 (3.3)	143 (4.0)	823 (3.2)	29 (2.4)	698 (3.4)
Born term, with normal birthweight (2500–4000 g)	18,558 (60.9)	2038 (56.9)	15,804 (61.5)	716 (60.0)	12,734 (61.8)
Born term, with macrosomia (>4000 g)	3029 (9.9)	355 (9.9)	2545 (9.9)	129 (10.8)	1965 (9.5)
Family history of diabetes, n (%)	9427 (30.9)	1288 (35.9)	7772 (30.2)	367 (30.7)	6432 (31.2)
Family history of other metabolic conditions ^d n (%)	19,904 (65.3)	2483 (69.3)	16,704 (65.0)	717 (60.1)	13,454 (65.3)
Time to cycle regularity, n (%)					
<1 year	12,057 (39.6)	1414 (39.5)	10,296 (40.1)	347 (29.1)	8211 (39.9)
1–2 years	5432 (17.8)	628 (17.5)	4611 (17.9)	193 (16.2)	3716 (18.0)
3–4 years	1543 (5.1)	212 (5.9)	1255 (4.9)	76 (6.4)	1042 (5.1)
>5 years	1703 (5.6)	207 (5.8)	1363 (5.3)	133 (11.1)	1148 (5.6)
Regular after hormones	3146 (10.3)	333 (9.3)	2625 (10.2)	188 (15.7)	2062 (10.0)
Not yet regular	2237 (7.3)	330 (9.2)	1747 (6.8)	160 (13.4)	1433 (7.0)
Age at first pregnancy					
Mean $\pm$ SD	25.0 $\pm$ 5.5	23.9 $\pm$ 5.2	25.1 $\pm$ 5.5	25.3 $\pm$ 5.5	25.3 $\pm$ 5.4
Median (IQR)	24 (20–29)	22 (20–27)	24 (20–29)	24 (21–29)	24 (21–29)
GDM at first pregnancy, n (%)	1261 (4.1)	196 (5.5)	1014 (3.9)	51 (4.3)	1209 (5.9)
GDM recurrence information, among N = 17,512 participants with two or more eligible pregnancies ^b					
GDM occurrence category, n (%)					
GDM single occurrence	1460 (8.3)	184 (9.6)	1222 (8.2)	54 (7.8)	/
Recurrent GDM	652 (3.7)	101 (5.3)	522 (3.5)	29 (4.2)	/
No GDM	15,400 (87.9)	1634 (85.1)	13,157 (88.3)	609 (88.0)	/
GDM recurrence rate among those with GDM at first pregnancy, n (%)	386 (53.6)	64 (58.7)	303 (52.0)	19 (67.9)	/
Among a subset of participants who provided early-life weight and height data (N = 7614):					
N	7614	767	6528	319	5115
BMI z-score at menarche, mean $\pm$ SD	0.32 $\pm$ 1.24	1.00 $\pm$ 1.40	0.28 $\pm$ 1.17	−0.49 $\pm$ 1.43	0.32 $\pm$ 1.25



TABLE 1 (Continued)

Baseline characteristics ^a	Overall	By age at menarche, years old			Those whose 1st pregnancy was ≥ 24 weeks with a live birth ^b
		<11	11–15	≥ 16	
BMI at menarche, <i>n</i> (%)					
Healthy/underweight	5678 (74.6)	348 (45.4)	5078 (77.3)	282 (88.4)	3808 (74.4)
Overweight	1176 (15.4)	211 (27.5)	948 (14.5)	17 (5.3)	793 (15.5)
Obese	760 (10.0)	208 (27.1)	532 (8.1)	20 (6.3)	514 (10.0)

Abbreviations: GDM, gestational diabetes mellitus; IQR, interquartile range; SD, standard deviation.

^aAll characteristics are based on participant self-report. Numbers may not add to total *N* (and percentages may not add to 100%) due to missingness.

^bEligible pregnancies defined in the main analysis: pregnancies that lasted ≥ 24 weeks with a live birth.

^c'Other' included: American Indian or Alaska Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander, or none of these categories can fully describe the participant. Multiple races correspond to those who selected more than one race/ethnicity category.

^dOther metabolic conditions include obesity, high cholesterol or hypertension.

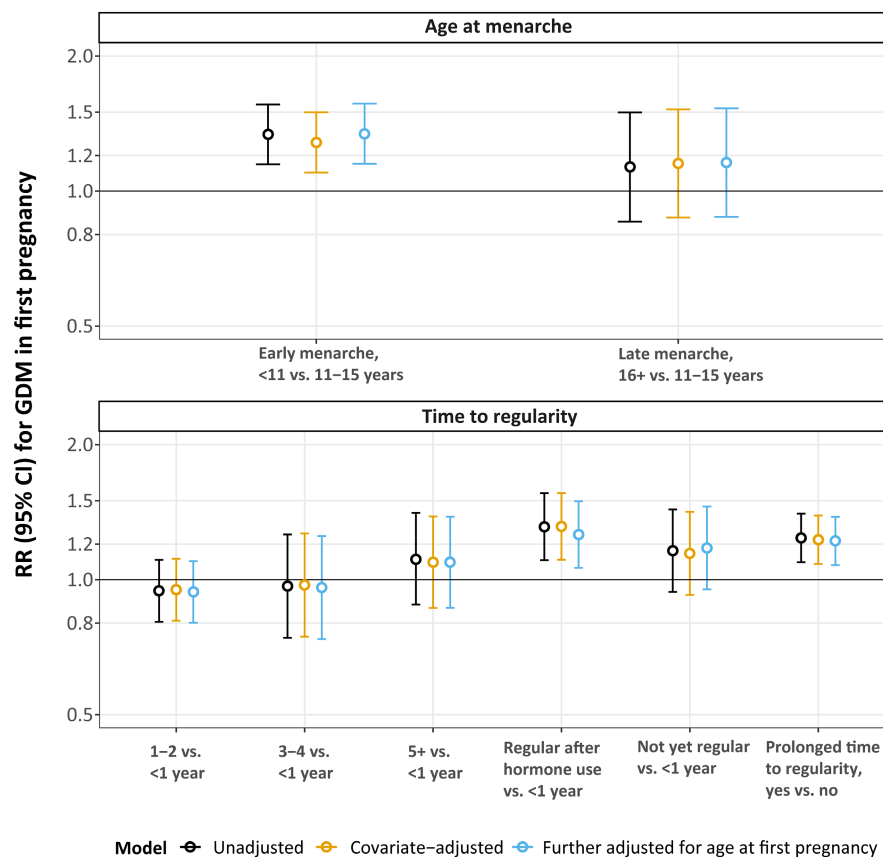


FIGURE 1 Associations of age at menarche and time-to-regularity with the risk of GDM in first pregnancy from modified Poisson regression models, among 20,591 participants whose first pregnancy lasted ≥ 24 weeks with a live birth, missing values imputed by MICE. CI, confidence interval; GDM, gestational diabetes; MICE, multiple imputation by chained equations; RR, risk ratio. Prolonged time to regularity defined as did not spontaneously establish regularity within 4 years since menarche (i.e. 5+ years, not yet regular or regular after hormone use). Covariate-adjusted model: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes and family history of other metabolic conditions. Model further adjusted age at first pregnancy: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes, family history of other metabolic conditions and age at first pregnancy.

Figure 1 shows the associations of age at menarche and time-to-regularity with GDM in first pregnancy (effect estimates in Table S2). When adjusted for covariates and age at first pregnancy, early

menarche (<11 vs. 12–15 years) is associated with a 1.3-times higher risk of GDM in first pregnancy. The associations for age at menarche in years appear non-linear from a GAM (Figure S4). Prolonged

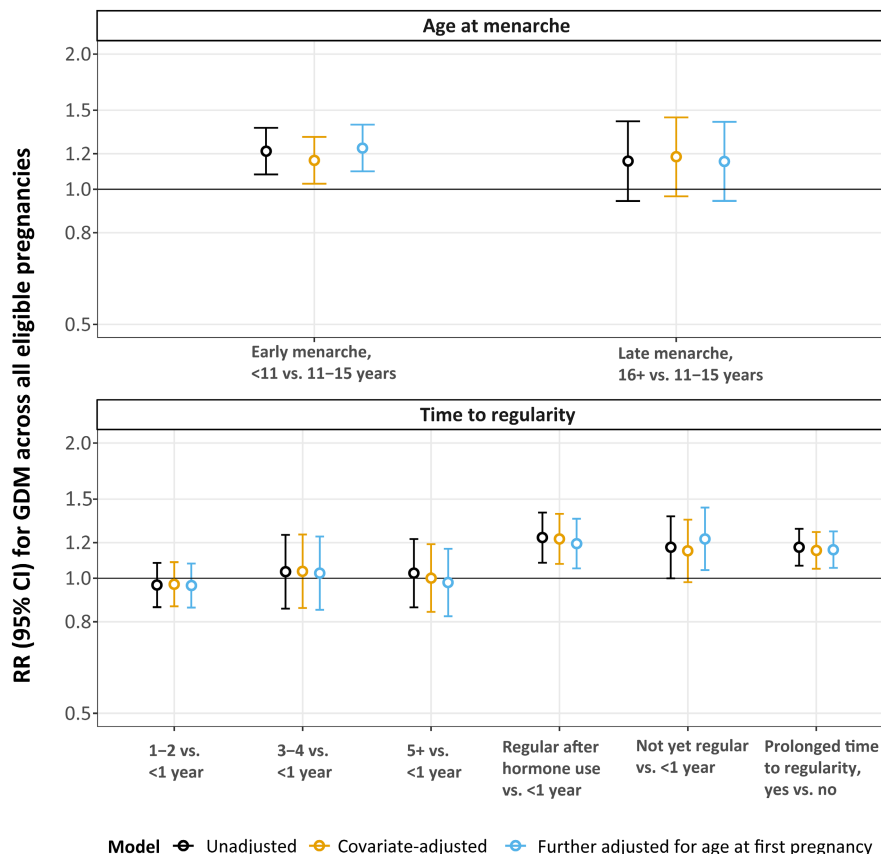


FIGURE 2 Associations of age at menarche and time-to-regularity with the risk of GDM across all eligible pregnancies from cluster-weighted Poisson GEE models, among 26,091 participants with at least one eligible pregnancy, missing values imputed by MICE. CI, confidence interval; GDM, gestational diabetes; GEE, generalised estimating equations; MICE, multiple imputation by chained equations; RR, risk ratio. The range for number of eligible pregnancies per participant was 1–10 in this analysis. Prolonged time to regularity defined as did not spontaneously establish regularity within 4 years since menarche (i.e. 5+ years, not yet regular or regular after hormone use). Covariate-adjusted model: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes and family history of other metabolic conditions. Model further adjusted age at first pregnancy: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes, family history of other metabolic conditions and age at each pregnancy.

time-to-regularity is associated with a 1.2-times higher risk of GDM in first pregnancy. Compared to time-to-regularity <1 year, establishing regularity due to hormone use is associated with a 1.3-times higher risk of GDM in first pregnancy.

Figure 2 presents the associations of age at menarche and time-to-regularity with the risk of GDM across pregnancies from CW-GEE models, among 26,091 participants with ≥ 1 eligible pregnancy (effect estimates in Table S3). When adjusted for covariates and age at pregnancy, early menarche is associated with a 1.2-times higher risk of GDM across pregnancies. Prolonged time-to-regularity is associated with a 1.2-times higher risk of GDM across pregnancies. Compared to time-to-regularity <1 year, establishing regularity due to hormone use or never establishing regularity are associated with a 1.2-times higher risk of GDM across pregnancies.

Figure 3 shows the associations of age at menarche and time-to-regularity with GDM single occurrence or recurrence (vs. no GDM) from multinomial logistic regression, among 17,512 participants with ≥ 2 eligible pregnancies (effect estimates in Table S4).

Early menarche and prolonged time-to-regularity are associated with a 1.5- and 1.2-times higher odds of recurrent GDM, respectively.

Among 7614 participants who provided BMI at menarche, 25.4% were overweight or obese at menarche (Table 1). Figure 4 shows that the BMI-adjusted models yielded effect estimates further away from the null for the associations between early menarche and GDM in first pregnancy, but the magnitude of change in effect estimates is small. Similar patterns are observed when using MICE to impute BMI at menarche and other variables among all 20,591 participants, or when using IPW for missing BMI at menarche (Figure 4). For the associations between prolonged time-to-regularity and GDM in first pregnancy, adjusting for BMI at menarche had little impact on the estimates across all three approaches (Figure 4).

Using data from all pregnancies regardless of gestation length or outcome yielded similar estimates (Table S5). When using MELR models, the positive associations remained for age at menarche in years (Table S6). Further adjusting for calendar year at pregnancy (Table S7), restricting to singletons (98.1%) (Table S8), limiting information to ≤ 4 th

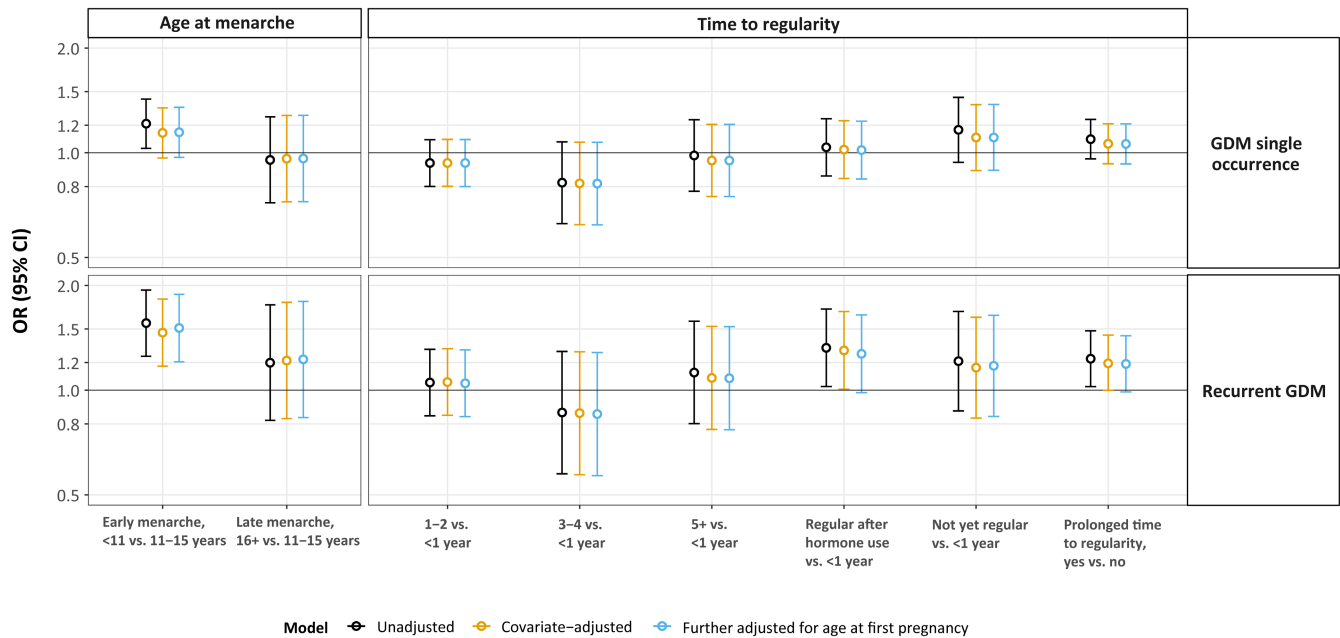


FIGURE 3 Associations of age at menarche and time-to-regularity with odds of GDM single occurrence or recurrence (vs. no GDM), among 17,512 participants with two or more eligible pregnancies, missing values imputed by MICE. CI, confidence interval; GDM, gestational diabetes; MICE, multiple imputation by chained equations; OR, odds ratio. The range for number of eligible pregnancies per participant was 2–10 in this analysis. Prolonged time to regularity defined as did not spontaneously establish regularity within 4 years since menarche (i.e. 5+ years, not yet regular or regular after hormone use). Covariate-adjusted model: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes and family history of other metabolic conditions. Model further adjusted age at first pregnancy: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes, family history of other metabolic conditions and age at first pregnancy.

pregnancy (Table S9 and Table S10) led to similar point estimates. There was no EM by PCOS on the additive scale (Figure S5). Among 2112 individuals who ever had GDM, early menarche was associated with higher risk of a subsequent GDM (Table S11). Removing current SES from the covariates led to similar estimates (Table S12). Those with very early menarche (<9 years) had the highest RR for GDM (Table S13).

4 | COMMENT

4.1 | Principal findings

In this large US cohort of participants with early-life menstrual characteristics, we found that earlier menarche and prolonged time-to-regularity are consistently associated with higher risk of GDM in first pregnancy, across pregnancies and GDM recurrence, independent of maternal age. Overall, our findings suggest that early-life menstrual cycle characteristics may serve as markers for GDM, and emphasise the need to explore childhood or adolescence as a potential time window for risk identification and prevention.

4.2 | Strengths of the study

Our study has several unique strengths. First, a large study size, accompanied by up to 10 pregnancies per individual, allowed for

sufficient statistical power to detect associations, not only for GDM in a single pregnancy, but also across multiple pregnancies and GDM recurrence. Second, we used statistical methods to account for within-individual correlations and informative cluster size. Third, we were able to account for several important early-life confounders such as birthweight, family history and early-life BMI. Lastly, our study evaluated the association between time from menarche to cycle regularity and GDM, identifying an additional early-life time window for intervention with this understudied hallmark of menstrual health.

4.3 | Limitations of the data

First, all variables were self-reported at enrolment, introducing potential recall bias. However, other validation studies showed high accuracy of recalled pregnancy history and menarche.⁴⁹ Our survey question of recalled age at menarche is similar to survey questions used in other studies with validated accuracy.⁵⁰ Furthermore, with the exposures (menarche and time-to-regularity) being asked at the beginning of the reproductive history survey, preceding the questions of pregnancy history, misclassification of exposures is likely to have been non differential with respect to GDM, leading to effect estimates biased towards the null, suggesting that our positive findings to be robust. Large epidemiologic studies also showed high accuracy of self-reported GDM diagnosis.^{51–53} Second, early-life BMI was only

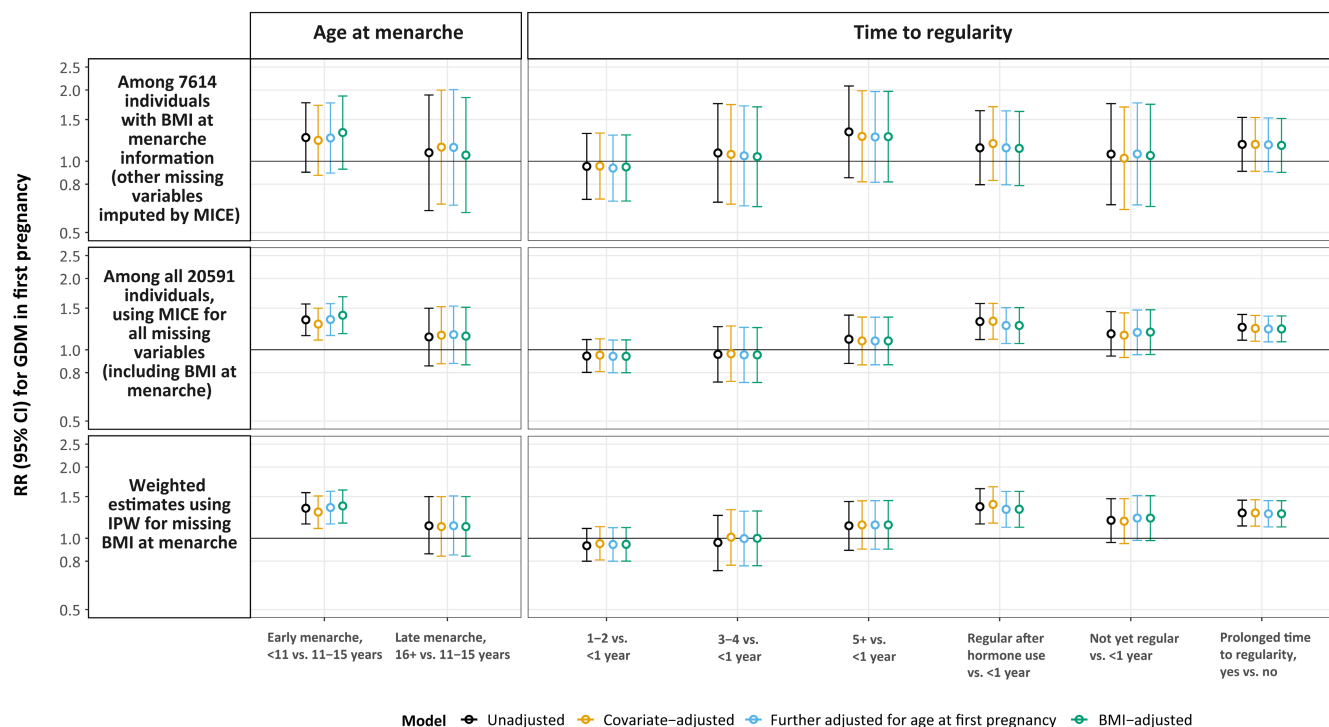


FIGURE 4 Associations of age at menarche and time-to-regularity with risk of GDM in first pregnancy, adjusted for BMI at menarche, comparing estimates from different groups and/or methods. CI, confidence interval; GDM, gestational diabetes; IPW, inverse probability weighting; MICE, multiple imputation by chained equations; RR, risk ratio. Prolonged time to regularity defined as did not spontaneously establish regularity within 4 years since menarche (i.e. 5+ years, not yet regular or regular after hormone use). Covariate-adjusted model: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes and family history of other metabolic conditions. Model further adjusted age at first pregnancy: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes, family history of other metabolic conditions and age at first pregnancy. BMI-adjusted model: Adjusted for race and ethnicity, socioeconomic status scale, participant's birthweight and preterm status, family history of diabetes, family history of other metabolic conditions, age at first pregnancy and BMI at menarche.

available among a subset of participants, and residual confounding due to inaccurate recall of early-life weight and height is possible. Third, we did not collect other early-life information, for example, in-utero epigenetic factors that may contribute to the pathophysiology of GDM and menarche.^{54,55} Fourth, we did not collect information such as GDM subtype or glucose measurements for past pregnancy history. Lastly, given the self-selection into the study and inclusion and exclusion criteria, our findings may not be fully generalizable to all US individuals who menstruate or to other populations.

4.4 | Interpretation

Our findings of an association of earlier menarche with higher GDM risk in the first pregnancy were consistent with some of the prior studies, while the examination of GDM risk across multiple pregnancies adds further information on overall and recurrence risk. Previous moderate-sized studies (2000–5000 individuals) showed mixed findings.^{11,15,56} Another large-sized US-based study ($n=27,482$) indicated a positive association between earlier menarche and GDM, but generalizability was limited by a predominantly White population who shared a common profession and

educational qualification.¹⁰ Early menarche is associated with altered oestrogen and SHBG levels that may impact insulin sensitivity⁹ or pre-pregnancy adiposity that may be on the causal pathway for GDM.³⁴ Interestingly, in our study, BMI at menarche did not appear to be a strong confounder. Further research on the potential role of adulthood adiposity, hormone levels and other lifestyle factors as multiple, potential mediators over the reproductive life course are needed. A Mendelian randomization study suggested genetic variants relevant to earlier menarche were independently associated with higher risk of GDM.⁵⁷ However, this was among individuals of European descent and may not be generalizable to a more heterogeneous population,²³ such as individuals of Asian descent who had much higher prevalence of GDM. Future research on racial and ethnic differences in early-life menstrual characteristics and GDM may further identify similar or distinct pathways through social, biological and environmental determinants. Our study reports positive associations between early menarche with all three GDM outcome measures: GDM in first pregnancy, across pregnancies and recurrent GDM. While some studies evaluated risk factors (mostly during the perinatal or adulthood windows) for recurrent GDM,⁷ evidence from population-based cohorts on early-life factors remain lacking. Our study in a large cohort presents

opportunities to identify individuals at high risk of recurrent GDM at an earlier time during the reproductive lifecourse.

As far as we are aware, our findings of associations between prolonged time-to-regularity and higher risk of GDM have not been previously reported. Prolonged time-to-regularity has been associated with adverse outcomes,⁵⁸ and may serve as an early-life vital sign for later-life health. Although some studies identified long or irregular menstrual cycles in mid-adulthood to be associated with higher GDM risk,^{15,17,59} no study has reported early-life menstrual characteristics and GDM risk. Previous studies suggest regular menstruation should be established within 1–2 years after menarche,⁶⁰ but evidence remains limited on whether delays beyond 2 years warrant clinical or lifestyle intervention. Long or irregular cycles is a hallmark of PCOS,⁴⁷ and prolonged time from menarche to regularity could be an early-life marker for PCOS. In our analysis, there was no effect medication by self-reported PCOS for the associations of menarche or time-to-regularity with GDM. This could be due to PCOS and menstrual cycle characteristics acting on the same biological pathway, but it is also possible that other factors involved in hormonal disruption and insulin resistance (e.g. BMI in adulthood, diet or lifestyle)⁴ may play larger roles, warranting future research. Our study suggests an additional time window during the post-menarche years that may inform GDM risk identification and intervention.

5 | CONCLUSIONS

In summary, earlier menarche and prolonged time-to-regularity are associated with higher risk of GDM in first pregnancy, across multiple pregnancies and GDM recurrence in a US cohort. This study provides evidence to further understanding of the markers for GDM and emphasises the need to explore childhood or adolescence as potential time windows for screening.

AUTHOR CONTRIBUTIONS

Z.W. was responsible for the study design, data preparation, statistical analyses, the interpretation of the results and manuscript writing. S.M. oversaw all aspects of the work and contributed to developing the concept, analysis plan, interpretation of the results, manuscript writing and revision. A.M.J., D.D.B. and A.J.W. provided critical reviews of the analysis and contributed to the interpretation of the results and manuscript revision. B.A.C., R.H., J.P.O. and M.A.W. contributed to the study design, interpretation of the results, manuscript writing and revision. C.L.C. and T.F.C. contributed to review of content in relation to the publication policy of Apple Inc. and did not participate in the analysis and interpretation of data. All the authors critically discussed the findings and provided feedback for the final draft. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

C.L.C. and T.F.C. own Apple Inc. stock and are employed by Apple Inc. Other co-authors have no conflict of interest. There were no financial relationships with any organisations that might have an interest in the submitted work in the previous 3 years and no other relationships or activities that could appear to have influenced the submitted work.

DATA AVAILABILITY STATEMENT

Aggregated data that support the findings of this study may be available upon reasonable request from the corresponding author and senior author. Any request for data will be evaluated and responded to in a manner consistent with policies intended to protect participant confidentiality and language in the Study protocol and informed consent form.

PATIENT CONSENT STATEMENT

All participants provided informed consent at enrolment.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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